

What's Different in SaaS

Series: M2S | **Notebook:** 1 of 8 | **Created:** January 2026 | **Last Updated:** 02/06/2026

Congratulations on your new Dynatrace SaaS tenant! This notebook series guides you through migrating your monitoring from Managed to SaaS.

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Prerequisites

Before starting this series, ensure you have:

Requirement	Description
SaaS tenant provisioned	Your new Dynatrace SaaS environment is ready
Managed environment access	Administrator access to your existing Managed deployment
API Tokens	Tokens with <code>entities.read</code> , <code>settings.read</code> scopes on both environments

Learning Objectives

By the end of this notebook, you will:

- Understand the Grail architecture that powers your new SaaS tenant
 - Know the key differences between Managed and SaaS
 - Have a complete assessment of your Managed environment
 - Be ready to plan your migration strategy
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1. Introduction

What This Series Covers

Notebook	Focus
M2S-01 (this notebook)	What's different in SaaS
M2S-02	Migration framework overview
M2S-03	Planning and assessment
M2S-04	Architecture and design
M2S-05	Configuration migration
M2S-06	OneAgent and ActiveGate migration
M2S-07	Security and privacy
M2S-08	Validation and optimization

Important: Your SaaS Tenant Uses Grail

Your new SaaS tenant is powered by **Grail**, Dynatrace's modern data lakehouse architecture. This means:

- Some features work differently than in Managed
- New capabilities are available that weren't in Managed
- Some legacy features are not available

This notebook helps you understand these differences so you can plan accordingly.

2. Grail Architecture Overview

What is Grail?

Grail is Dynatrace's unified data lakehouse that powers all data storage and analytics in SaaS. It replaces the traditional time-series database used in Managed.

Grail vs Legacy Architecture

Aspect	Legacy/Managed	Grail (SaaS)
Data storage	Time-series database	Unified data lakehouse
Query language	USQL (limited)	DQL (full-featured)
Dashboards	Classic dashboards	Modern dashboards + Notebooks
Log analytics	Log Monitoring v1	Log Management powered by Grail
Business events	Limited	Full business analytics
Retention	Fixed tiers	Flexible, cost-optimized

Grail-Exclusive Features

These capabilities are only available on your new SaaS tenant:

Feature	Description
Grail	Petabyte-scale data lakehouse
DQL	Context-aware queries across all data types
Notebooks	Interactive analysis with collaboration
Automations	Advanced workflow engine
OpenPipeline	Custom data processing and routing
Business Analytics	Full business context and KPIs

Dynatrace SaaS Benefits

Reduced Operations

No server maintenance
No patching or upgrades

Automatic Updates

Always on latest version
New features immediately

Elastic Scalability

Scale automatically
No capacity planning

Enhanced Security

Dynatrace manages infra
SOC2, ISO certified

Global Availability

Multi-region deployment
Built-in resilience

Cost Optimization

Predictable licensing
No hardware costs

Migration Impact Summary

Managed (Before)
Infrastructure overhead + Manual updates + Capacity planning

→

SaaS (After)
Zero ops + Auto updates + Elastic scale + Latest features

3. Key Differences from Managed

What Changes

Area	Managed	SaaS (Grail)
Queries	USQL	DQL (required for new features)
Dashboards	Classic only	Modern dashboards (recommended)
Log Management	Log Monitoring v1	Log Management via Grail
Metrics	Classic metrics	Grail-powered metrics
Alerting	Classic alerts	Workflows and Davis Analyzers

What Stays the Same

Your existing investments are preserved:

- OneAgent deployment patterns
- Monitoring configurations (with migration)
- DQL query skills (DQL works on both)
- Classic dashboards (still supported, modern recommended)
- Core alerting concepts

What You'll Need to Learn

Topic	Why
DQL	Required for querying Grail data
Modern dashboards	New visualization approach
Workflows	Replaced classic alerting profiles
OpenPipeline	New log processing model
Notebooks	Interactive analysis tool

Important Limitations to Know

Limitation	Impact	Planning Note
Historic data	Cannot be migrated	Plan for baseline period in SaaS
Host limit	25,000 per tenant	Split tenants if exceeding
SAML signing	IdP must sign full message	Verify IdP configuration
OneAgent versions	9-12 month support window	Check oldest versions

SAML/SSO Requirements

 **Critical:** Your Identity Provider (IdP) must sign the **entire SAML message**, not just the assertion. Azure AD meets this requirement by default.

IdP Consideration	Requirement
SAML message signing	Full message, not just assertion
Group limit (Azure Entra)	150 groups per user in SAML claim
Group filtering	Filter to Dynatrace-related groups only

4. Assessing Your Managed Environment

Before planning your migration, document your current state.

Important: DQL (Grail) is only available in SaaS environments. For Managed environments, use the **Entities API v2** or the **Dynatrace UI** to gather this information.

Environment Size Assessment

Use the Entities API v2 to gather counts from your Managed environment:

```
# Count total monitored hosts
curl -X GET "https://{your-managed-url}/api/v2/entities?
entitySelector=type(HOST)&pageSize=1" \
  -H "Authorization: Api-Token {TOKEN}" | jq '.totalCount'

# Count hosts by OS type
curl -X GET "https://{your-managed-url}/api/v2/entities?
entitySelector=type(HOST)&fields=properties.osType" \
  -H "Authorization: Api-Token {TOKEN}"

# Count services
curl -X GET "https://{your-managed-url}/api/v2/entities?
entitySelector=type(SERVICE)&pageSize=1" \
  -H "Authorization: Api-Token {TOKEN}" | jq '.totalCount'

# Count applications
curl -X GET "https://{your-managed-url}/api/v2/entities?
entitySelector=type(APPLICATION)&pageSize=1" \
  -H "Authorization: Api-Token {TOKEN}" | jq '.totalCount'
```

Or use the Dynatrace UI:

- **Hosts:** Navigate to Infrastructure → Hosts (count shown in header)
- **Services:** Navigate to Applications & Microservices → Services
- **Applications:** Navigate to Applications & Microservices → Frontend

OneAgent Version Assessment

Understanding your OneAgent versions helps plan the migration. Use the Entities API:

```
# Get OneAgent versions
curl -X GET "https://{your-managed-url}/api/v2/entities?
entitySelector=type(HOST)&fields=properties.oneAgentVersion" \
  -H "Authorization: Api-Token {TOKEN}"
```

Or use the Dynatrace UI: Navigate to **Manage → Deployment status → OneAgent**

ActiveGate Assessment

Identify your ActiveGate deployment using the Entities API:

```
# Get ActiveGate inventory
curl -X GET "https://{your-managed-url}/api/v2/entities?
entitySelector=type(ENVIRONMENT_ACTIVE_GATE)" \
-H "Authorization: Api-Token {TOKEN}"
```

Or use the Dynatrace UI: Navigate to **Manage** → **Deployment status** → **ActiveGates**

```
// Check OneAgent versions across hosts
fetch dt.entity.host
| summarize hostCount = count(), by:{installerVersion}
| sort hostCount desc
```

5. Next Steps

The SaaS Upgrade Assistant

Dynatrace provides the [SaaS Upgrade Assistant](#) app to automate your migration from Managed to SaaS. This is the **recommended primary tool** for most migrations.

Feature	Description
Automated Configuration Import	Export configurations from Managed and upload to SaaS—most settings and dashboards migrate automatically
Selective Import	Smart dependency-aware import lets you choose which configurations to deploy per wave
Dashboard Ownership Migration	Automatically updates dashboard owners from Managed to SaaS user identifiers
Entity ID Adjustment	Handles entity ID changes between environments automatically
Progress Tracking	Real-time upgrade status with deployment result downloads (CSV)
Bulk Editing	Edit and correct failed configurations individually or in bulk mode
Preview Changes	Review all changes before deploying to ensure accuracy

How It Works

1. **Export** configuration from your Managed Cluster Management Console
2. **Upload** the configuration archive to the SaaS Upgrade Assistant app in your target SaaS tenant
3. **Review** imported configurations—grouped by type with error highlighting
4. **Edit** any failed or incompatible configurations (single or bulk mode)
5. **Deploy** selected configurations to your SaaS environment
6. **Track** progress and download deployment result reports

Version Alignment Requirement

Important: For best results, align your Managed cluster and SaaS environment to the **same major version** (e.g., both 1.294.x). Mismatched versions can cause false-positive migration failures. The SaaS Upgrade Assistant requires **Dynatrace Managed version 1.294 or later**.

IAM Requirements

Users need the `upgrade-assistant:environments:write` IAM policy assigned in Account Management to use the SaaS Upgrade Assistant.

Tip: See the full documentation at [SaaS Upgrade Assistant](#) and contact your Dynatrace account team for guided migration support.

Migration Tooling Options

Tool	Best For	Approach
SaaS Upgrade Assistant	Most migrations—guided UI-based migration	Export from Managed, import via app
Monaco	Configuration-as-code standardized deployments	YAML-based config management
Terraform	Infrastructure-as-code environments	Dynatrace Terraform provider
Settings API	Custom automation scripts	Direct API export/import

Warning: Avoid mixing tooling approaches. For example, Monaco YAML templates can conflict with the SaaS Upgrade Assistant. Choose one primary method for consistency.

Immediate Actions

1. **Run the assessment queries** - Document your Managed environment size
2. **Verify network connectivity** - Can you reach your SaaS tenant?
3. **Create API tokens** - On both Managed and SaaS
4. **Install the SaaS Upgrade Assistant** - On your target SaaS tenant
5. **Review the checklist** - Identify any gaps

Continue the Series

Next Notebook	Focus
M2S-02: Migration Framework	Understanding the 3-phase migration approach

Additional Resources

- [SaaS Upgrade Assistant Documentation](#)
- [SaaS Upgrade Assistant on Dynatrace Hub](#)
- [Upgrading from Dynatrace Managed to SaaS](#)

- [Dynatrace SaaS Documentation](#)
 - [Grail Data Lakehouse](#)
 - [DQL Reference](#)
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Summary

In this notebook, you learned:

- How Grail architecture differs from Managed
- Key changes to expect in your SaaS tenant
- Important limitations and requirements
- How to assess your current Managed environment
- The SaaS Upgrade Assistant as the primary migration tool
- Available migration tooling options

Key Takeaway: Your SaaS tenant uses Grail, which means new capabilities but also some changes in how you work. The [SaaS Upgrade Assistant](#) automates the majority of configuration migration—start by installing it on your SaaS tenant.

Continue to M2S-02: Migration Framework Overview to learn about the structured approach to migration.